Proof.

We know the sine formula: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R$

Now, $\dfrac{a-b}{a+b}\cot\dfrac{C}{2} = \dfrac{2R\sin A - 2R\sin B}{2R\sin A + 2R\sin B}\cot\dfrac{C}{2}$

$$= \frac{\sin A - \sin B}{\sin A + \sin B}\cot\frac{C}{2}$$

$$= \frac{2\cos\dfrac{A+B}{2}\sin\dfrac{A-B}{2}}{2\sin\dfrac{A+B}{2}\cos\dfrac{A-B}{2}}\cot\frac{C}{2}$$

$$= \cot\frac{A+B}{2}\tan\frac{A-B}{2}\cot\frac{C}{2}$$

$$= \cot\left(90^\circ - \frac{C}{2}\right)\tan\frac{A-B}{2}\cot\frac{C}{2}$$

$$= \tan\frac{C}{2}\tan\frac{A-B}{2}\cot\frac{C}{2} = \tan\frac{A-B}{2}$$

Similarly we can prove the other two results.

3.7.2 Law of Cosines

When two sides and included angle or the three sides of a triangle are given, the triangle cannot be solved using the sine formula. In such a situation, the law of cosines can be used to solve the triangle. Also, the Law of Cosines is used to derive a formula for finding the area of a triangle given two sides and the included angle.

Theorem 3.3 (The Law of Cosines)**:** In $\triangle ABC$, we have

$$\cos A = \frac{b^2+c^2-a^2}{2bc};\ \cos B = \frac{c^2+a^2-b^2}{2ca};\ \cos C = \frac{a^2+b^2-c^2}{2ab}.$$

Proof. In $\triangle ABC$, draw $AD \perp BC$.

In $\triangle ABD$, we have $AB^2 = AD^2 + BD^2 \Rightarrow c^2 = AD^2 + BD^2$.

Now, we find the values of AD and BD in terms of the elements of $\triangle ABC$.

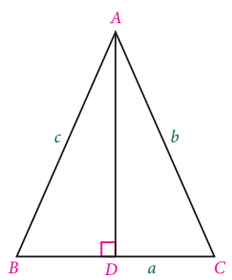


Figure 3.19

$$\frac{AD}{AC} = \sin C \Rightarrow AD = b\sin C$$

$$BD = BC - DC = a - b\cos C$$

$$c^2 = (b\sin C)^2 + (a - b\cos C)^2$$

$$= b^2\sin^2 C + a^2 + b^2\cos^2 C - 2ab\cos C$$

$$= a^2 + b^2\left(\sin^2 C + \cos^2 C\right) - 2ab\cos C$$

$$= a^2 + b^2 - 2ab\cos C$$

Thus, $c^2 = a^2 + b^2 - 2ab\cos C$ or $\cos C = \dfrac{a^2+b^2-c^2}{2ab}$.

Similarly, we can prove the other two results, namely

$$a^2 = b^2 + c^2 - 2bc\cos A \text{ or } \cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$b^2 = c^2 + a^2 - 2ca\cos B \text{ or } \cos B = \frac{c^2+a^2-b^2}{2ca}$$